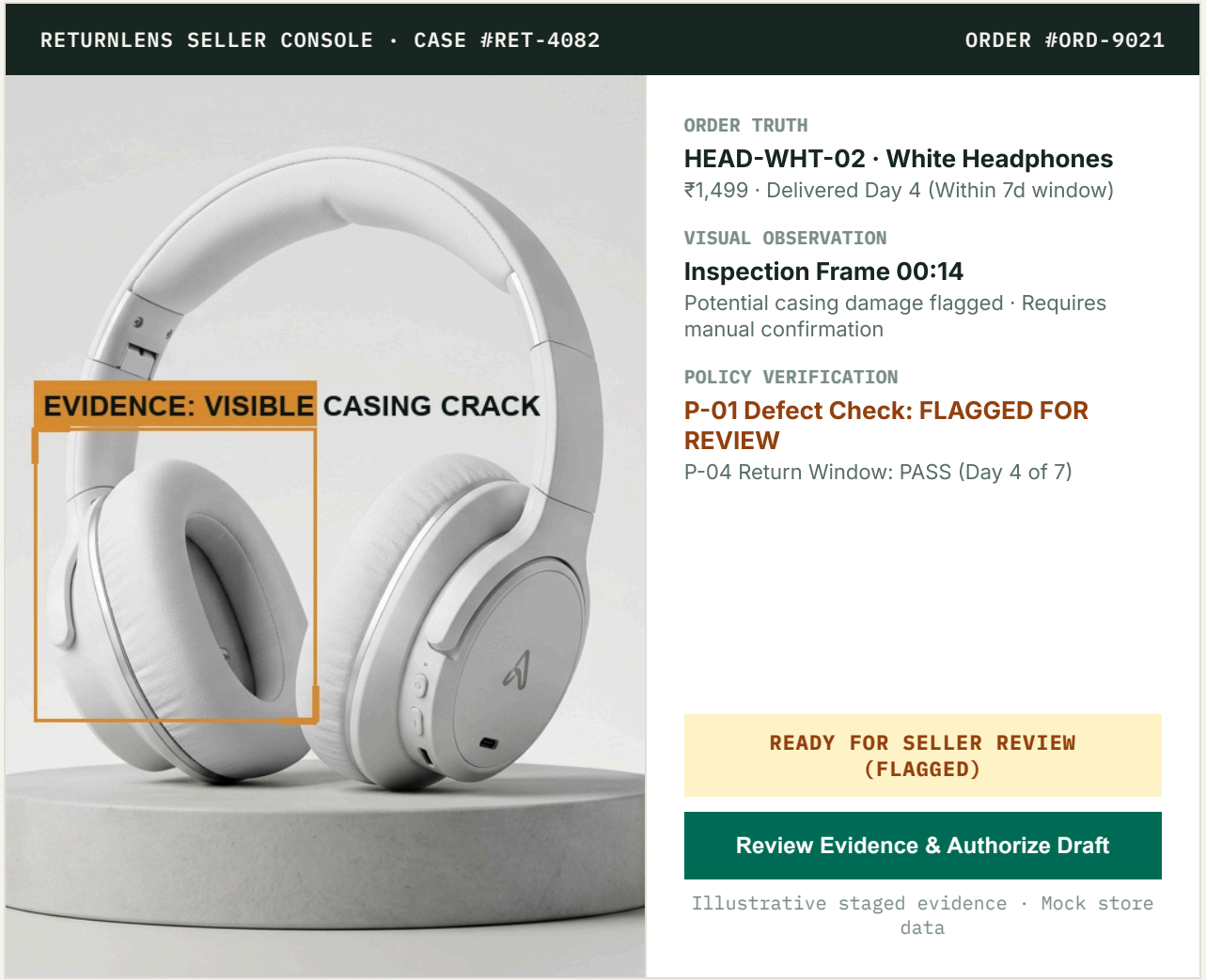
AGENTX HACKATHON 2026 · TRACK PS-04

RETURNLENS

“Review returns with cited video evidence.”

An evidence-backed return triage agent for small direct-to-consumer sellers.



Team Rookie · 3 members · PS-04 Track
Target User: Solo D2C merchant managing orders and returns alone.
Scope in 24h: Return triage core + 1 zero-vision order-status route.
Guardrails: Fail-closed architecture · No autonomous refunds.

THE PROBLEM · SOLO SELLER BOTTLENECK

One return decision. Four disconnected sources.

The return decision is made from memory, before the evidence is checked.

01 BUYER CHAT

“Received damaged item”

Buyer upload: 30–60s unboxing video + reason in support message

02 ORDER RECORD

White Headphones · ₹1,499

Shopify mock store API product inventory catalog

CRITICAL INFORMATION GAP

NO SHARED EVIDENCE RECORD

Today the seller opens 4 tabs, then decides alone. A wrong refund loses inventory; an unfair denial loses a customer.

The gap is that no single record ties order truth, video frames, policy clauses, and decisions together.

03 COURIER TRACKING

Delivered 4 days ago

Shippo mock tracking API scan timestamp verification

04 RETURN POLICY

7-Day Window · Sealed Tag

Deterministic code rules P-01 to P-08, never prompts

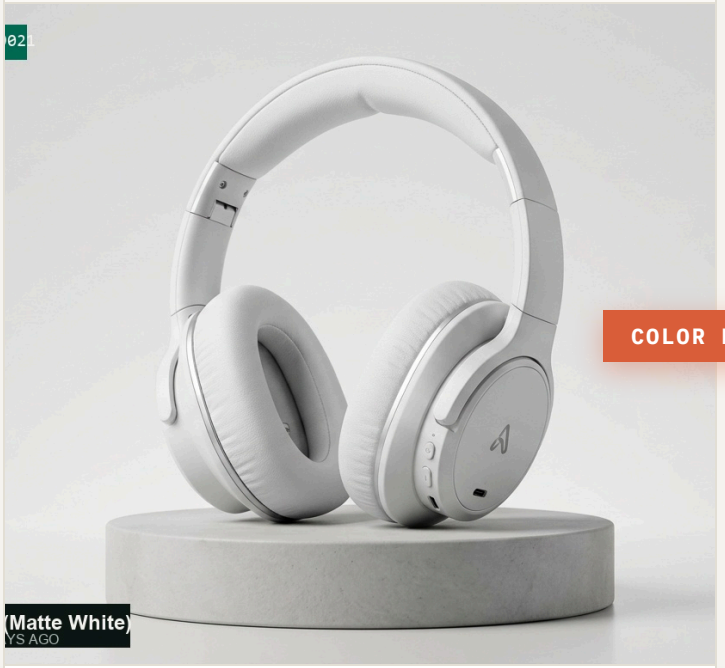
Fraud estimates differ by definition: 9% (NRF 2025, fraud only) vs 15.14% (Appriss/Deloitte 2024, fraud + abuse).

VISUAL EVIDENCE BENCHMARK · CORE DEMONSTRATION

The order says white. The video shows black.

ReturnLens surfaces the contradiction instead of auto-denying.

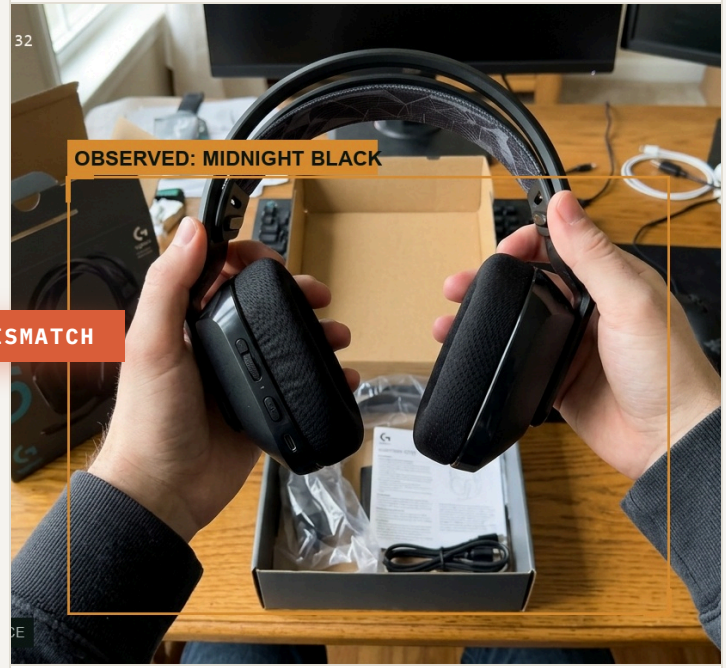
ORDER RECORD · STORE GROUND TRUTH



HEAD-WHT-02 · Matte White · ₹1,499
Delivered 4 days ago · Original tag present

COLOR MISMATCH

VIDEO FRAME · INTAKE TIMESTAMP 00:07



Observed Color: Midnight Black
Exact SKU: Not verified · Tag: Not visible in cited frame

DETERMINISTIC POLICY GATE

P-03 Check: COLOR MISMATCH

Specification contradiction flagged against Shopify order manifest.

```
typed JSON, requires_seller_approval: true
```

STATUS: ESCALATE (SELLER REVIEW REQUIRED)
Color mismatch observed · Exact SKU unconfirmed · No auto-denial.

SELLER ACTION OPTIONS

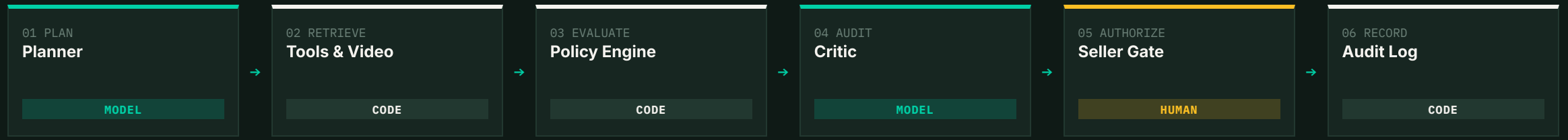
1. Accept as merchant mis-shipment (seller mistake)
2. Ask customer for outside-box shipping label (buyer mistake)
3. Hold return for physical inspection at warehouse (potential swap)

Deterministic Guardrails:
• Cap: 0 < amount ≤ min(₹4,999, remaining refundable amount, order paid amount)
• No automatic customer denial · No refund released without seller click

SYSTEM ARCHITECTURE · DETERMINISTIC PIPELINE

Models observe. Code verifies. Seller authorizes.

Policy checks and seller approval control actions.



SECONDARY PS-04 ROUTE **Order Status Inquiry:** Customer: “Where is my order?” → Mock Store API → Courier Scan → Instant Delivery Timestamp — Zero-Vision Fast Path

1. CONTRADICTION → ESCALATE
Discrepancy flagged (e.g. Case B). Inhibits automated label/refund. Seller reviews 3 options.

2. HARD FAIL → ESCALATE
Mandatory rule violation (e.g. expired return window). Seller makes final decision; never auto-deny.

3. UNCLEAR ONLY → ASK MORE
Low brightness or blurry barcode (e.g. Case C). Prompts exact re-film checklist (1 retry limit).

4. ALL PASS → APPROVE REVIEW
All clauses pass (e.g. Case A). Prepares mock label and drafts refund in pending_return_receipt.

- **Model Calls:** Exactly 3 base calls per case: planner, vision, critic, plus retries
- **Policy Clauses:** Deterministic Python code rules P-01 to P-08, never prompts
- **Critic Duty:** The critic reviews cited frames and flags unsupported observations
- **Safety Boundary:** Prompt injection cannot alter policy code or bypass human gate

DEMO SCENARIOS · GROUNDED VERIFICATION

Four scenarios. Four predictable behaviors.

Every decision can be clicked back to a frame, a clause, and a log entry.

CASE A · READY	**ORD-1042** (₹1,499) · SKU matches manifest, tag visible, defect confirmed within 7-day window.	→	Seller approval releases mock return label and refund draft in **pending_return_receipt**, not executed refund.
CASE B · MISMATCH	**HEAD-WHT-02** (White) ordered vs Frame 00:07 showing Black item · SKU unverified.	→	**ESCALATE** (no label, no refund, no denial). Seller options: mis-shipment / box label / physical inspection.
CASE C · UNCLEAR	**4-second dark video**, brightness 28, only 2 usable frames · Unclear evidence check.	→	Case stays open with exact re-film checklist. One retry limit prevents customer lock-out.
CASE D · CONTEXT	**Parcel-to-product take**: opening sealed parcel to product reveal in one continuous take.	→	Expected behavior: inspect sequence continuity and flag missing transitions for seller review.

EXECUTION PLAN · TEAM ROOKIE (3 MEMBERS)

Built for a 24-hour sprint.

Buildable in 24 hours by 3 people, if the scope stays narrow.

00H – 06H

Mock APIs & Agent Core

Person A (Backend)
LangGraph skeleton, mock store/courier fixtures, order status route

06H – 12H

Evidence Pipeline & Policy

Person B (Video & UI)
FFmpeg keyframe indexer, Python rules P-01 to P-08, critic prompt

12H – 18H

Integration & 4 Scenarios

Person C (Integration)
End-to-end wiring, 4 staged demo cases (A–D), seller console UI

18H – 24H

Testing, Polish & Submission

All 3 Members
Fallback testing, edge cases, video walkthrough recording, README

NEVER CUT (CORE ENGINE)

Deterministic policy engine (P-01 to P-08) · Evidence-pack frame citation · Fail-closed seller approval gate · Structured JSON audit log · 4 reproducible demo cases.

CUT FIRST (DEFERRED EXPANSION)

Video OCR barcode scanner · Auto-drafted customer email reply · Seller analytics dashboard · Production payment gateways / live WhatsApp webhooks.

"The winning hackathon agent is not the one with the most tool definitions. It is the one where every action is held to evidence, and the seller stays in control."